IE540
L11. Policy based RL : Deterministic PG, CEM

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Policy based RL (Policy Optimization)

- **Policy Gradient**
  - Stochastic policy gradient
    - REINFORCE
    - Actor-Critic : Stochastic policy (actor) & Value function (critic)
      - Q A-C : Critic is Q(s,a)
      - Advantage A-C (A2C), Async. Adv. A-C (A3C, Deepmind, ICML2016)
      - Other variants : TD A-C, TD(λ) A-C, …
    - TRPO(ICML 2015), PPO(arXiv 2017), PPG(arXiv 2020) : Schulman
    - SAC (Haarnoja, ICML 2018)
  - Deterministic policy gradient
    - DPG (Silver, ICML 2014)
    - DDPG (Deepmind, ICLR 2016)
    - TD3 (Fujimoto, ICML 2018)
- **Derivative Free Optimization**
  - Evolutionary strategy (ES) : population based
    - CEM (Cross-Entropy Method)
    - GA (Genetic Algorithm), SA (Simulated Annealing)
- **PG + ES** : Combine PG with ES
  - ERL (NIPS 2018) : GA + DDPG
  - CEM-RL (ICLR 2019) : CEM + TD3
  - …

Review: Stochastic Policy Gradient

$$d^{\pi_\theta}(s) = \frac{\rho^{\pi_\theta}(s)}{\sum_{x \in \mathcal{S}} \rho^{\pi_\theta}(x)}$$

$$\rho^{\pi_\theta}(s) = \sum_{s_0} p_0(s_0) \sum_{k=0}^{\infty} \gamma^k \, p(s_0 \to s, k, \pi_\theta)$$

➢ Stochastic Policy Gradient Theorem

- $J(\pi_\theta) = \mathbb{E}_{s \sim p_0, a \sim \pi_\theta}[Q^{\pi_\theta}(s,a)] \propto \mathbb{E}_{s \sim d^{\pi_\theta}, a \sim \pi_\theta}[Q^{\pi_\theta}(s,a)]$
  - $J(\pi_\theta) = \int_{\mathcal{S}} \rho^{\pi_\theta}(s) \int_{\mathcal{A}} \pi_\theta(a|s) Q^{\pi_\theta}(s,a) \, da \, ds$
- $\nabla_\theta J(\pi_\theta) = \int_{\mathcal{S}} \rho^{\pi_\theta}(s) \int_{\mathcal{A}} \nabla_\theta \pi_\theta(a|s) Q^{\pi_\theta}(s,a) \, da \, ds$

$$\propto \mathbb{E}_{s \sim d^{\pi_\theta}, a \sim \pi_\theta}[\nabla_\theta \log \pi_\theta(a|s) \, Q^{\pi_\theta}(s,a)]$$

Note

$$\nabla_\theta \log \pi_\theta(a|s) = \frac{\nabla_\theta \pi_\theta(a|s)}{\pi_\theta(a|s)}$$

  - Expectation form → sample based estimation
  - Though d^{π_θ} and $Q^{\pi_\theta}(s,a)$ depend on θ, $\nabla_\theta d^{\pi_\theta}$ or $\nabla_\theta Q^{\pi_\theta}(s,a)$ is not required

- How to estimate $Q^{\pi_\theta}(s,a)$?
  - REINFORCE (MC) : use $G_t = \sum_{k=t}^{\infty} \gamma^{k-t} r_t$ or $A_t = G_t - V_v(s_t)$
  - Vanilla A-C : Use critic function $Q_w(s,a)$
  - TD Adv. A-C : Use 1-step TD advantage → $\delta_t^V = r_t + \gamma V_v(s_{t+1}) - V_v(s_t)$
  - GAE $A_t = \sum_{k=0}^{\infty} (\gamma\lambda)^k \, \delta_{t+k}^V$

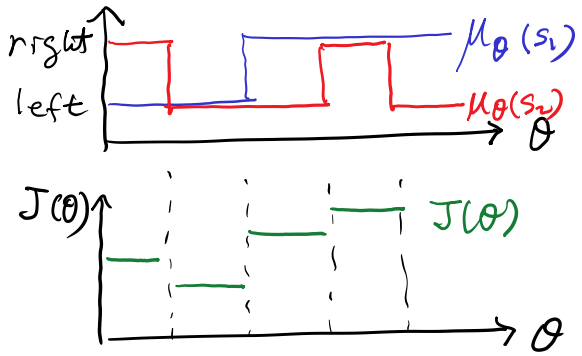
Policy Gradient

TAQ What would actually happen if SPG is used when optimal policy is deterministic (discrete action)?

- Continuous action space
  - such as robotic control tasks
  - cartpole? need modification
- Stochastic policy gradient : applicable to continuous action
  - Assumes stochastic policy $\pi_\theta(a|s)$: parametric function of θ
    - Discrete action : softmax policy
    - Continuous action : Gaussian policy
  - Policy gradient $\nabla_\theta J(\theta) \propto \mathbb{E}_{s\sim d_\theta, a\sim\pi_\theta}[\nabla_\theta \ln \pi_\theta(a|s)\, Q^{\pi_\theta}(s,a)]$: as an expectation form
  - Low performance for continuous action
    - especially if action space dim. is high

TAQ Why is SPG performance low?

- What if the desired policy is deterministic $a = \mu_\theta(s)$?
  - We want to find $\mu_\theta(s)$ s.t. $\max_{a\in\mathcal{A}} Q(s,a) \approx Q\big(s,\mu_\theta(s)\big)$?
  - Can we use gradient to update deterministic policy?
    - Deterministic policy for discrete action : not applicable
      - Small change of policy parameter θ → no change in action
        - » Policy is piecewise constant w.r.t. θ
        - » Gradient is zero almost everywhere
        - » Policy is discontinuous → Can't swap differentiation and expectation
    - Continuous action : θ can be updated using DPG



Deterministic Policy Gradient Algorithms (Silver, ICML 2014)

- **MDP:** $< \mathcal{S}, \mathcal{A}, P, r, p_0, \gamma >$
  - ✓ $\mathcal{S}$: state set (discrete or continuous)
  - ✓ $\mathcal{A}$: action set (continuous)
  - ✓ $p(s'|s,a)$: state transition probability distribution,
  - ✓ $r(s,a)$: reward function, expected reward for action a at state s
  - ✓ $p_0(s_0)$: distribution of the initial state s_0
  - ✓ $\gamma \in (0,1]$: discount factor
- **Policy :** parameterized policy function
  - ✓ $\pi_\theta(a|s)$: stochastic policy, probability dist. of actions at state s
  - ✓ $\mu_\theta(s)$: deterministic policy $\mu_\theta: \mathcal{S} \rightarrow \mathcal{A}$
- **Return** (total discounted reward), **Value Functions**
  - $G_t = \sum_{k=t}^{\infty} \gamma^{k-t} r(s_k, a_k)$
  - $V^\mu(s) = \mathbb{E}_{\tau\sim\mu}[G_t|S_t = s] = Q^\mu(s, \mu(s))$ and $Q^\mu(s,a) = \mathbb{E}_{\tau\sim\mu}[G_t|S_t = s, A_t = a]$
- **Performance Metric**
  - $J(\mu) = \mathbb{E}[G_t] = \mathbb{E}_{s_0\sim p_0}[V^\mu(s_0)] = \mathbb{E}_{s_0\sim p_0}[Q^\mu(s_0, \mu(s_0))] = \int_S p_0(s_0)\, Q^\mu(s_0, \mu(s_0))\, ds_0$
    - $J(\mu) = \mathbb{E}_{s_0\sim p_0}[V^\mu(s_0)] \propto \mathbb{E}_{s\sim d^\mu}[r(s, \mu(s))] \propto \mathbb{E}_{s\sim d^\mu}[V^\mu(s)]$ (see A1)
    - $d^\mu(s)$: discounted stationary distribution (under policy μ)

Deterministic Policy Gradient (Intuitive overview)

- Find a deterministic policy μ_θ s.t. $\max_{a \in \mathcal{A}} Q^{\mu_\theta}(s,a) \approx Q^{\mu_\theta}\big(s, \mu_\theta(s)\big)$
- Generalized Policy Iteration
  - Policy evaluation
    - evaluate $Q^{\mu_\theta}(s,a)$ from samples
  - Policy improvement
    - $\mu_\theta(s) \leftarrow \arg\max_{a \in \mathcal{A}} Q^{\mu_\theta}(s,a)$
      - this can be very expensive if $\mathcal{A}$ is continuous
    - $\theta \leftarrow \theta + \alpha \nabla_\theta Q^{\mu_\theta}(s, \mu_\theta(s))$: on which s?
    - $\theta \leftarrow \theta + \alpha \nabla_\theta \mathbb{E}_{s \sim d^{\mu_\theta}} [Q^{\mu_\theta}(s, \mu_\theta(s))]$
      - $\mathbb{E}_{s \sim d^{\mu_\theta}} [Q^{\mu_\theta}(s, \mu_\theta(s))] = \int_s d^{\mu_\theta}(s)\, Q^{\mu_\theta}\big(s, \mu_\theta(s)\big) ds$
    - $\theta \leftarrow \theta + \alpha \mathbb{E}_{s \sim d^{\mu_\theta}} [\nabla_\theta Q^{\mu_\theta}(s, \mu_\theta(s))]$
    - $\theta \leftarrow \theta + \alpha \mathbb{E}_{s \sim d^{\mu_\theta}} \left[\nabla_{a=\mu_\theta(s)} Q^{\mu_\theta}(s,a) \nabla_\theta \mu_\theta(s)\right]$
      - is it OK to ignore the dependence of d^{μ_θ} on θ?
        - » [ ]
        - » deterministic policy gradient theorem tells!!

Deterministic Policy Gradient Theorem

$p_0(s)$: initial state distribution

- Performance Metric

J_0 J_{EV} J_{ER}

$$J(\mu_\theta) = \mathbb{E}_{s_0 \sim p_0}[Q^{\mu_\theta}(s_0, \mu_\theta(s_0))] \propto \mathbb{E}_{s \sim d^{\mu_\theta}}[Q^{\mu_\theta}(s, \mu_\theta(s))] \propto \mathbb{E}_{s \sim d^{\mu_\theta}}[r(s, \mu_\theta(s))] \quad \text{(see A1)}$$

$$J(\mu_\theta) \propto \int_{\mathcal{S}} d^{\mu_\theta}(s)\, Q^{\mu_\theta}\big(s, \mu_\theta(s)\big) ds = \mathbb{E}_{s \sim d^{\mu_\theta}}\big[Q^{\mu_\theta}\big(s, \mu_\theta(s)\big)\big]$$

TAQ Exact relation btn J_0, J_{EV} and J_{ER}?

- Deterministic Policy Gradient Theorem (see A2)

$$\nabla_\theta J(\mu_\theta) = \int_{\mathcal{S}} \rho^{\mu_\theta}(s)\, \nabla_\theta Q^{\mu_\theta}\big(s, \mu_\theta(s)\big) ds = \int_{\mathcal{S}} \rho^{\mu_\theta}(s)\, \nabla_\theta \mu_\theta(s) \nabla_{a=\mu_\theta(s)} Q^{\mu_\theta}(s, a)\, ds$$

$$\propto \mathbb{E}_{s \sim d^{\mu_\theta}}\big[\nabla_\theta \mu_\theta(s) \nabla_{a=\mu_\theta(s)} Q^{\mu_\theta}(s, a)\big]$$

- $p_0(s)$: initial state distribution
- $\rho^{\mu_\theta}(s_0 \to s) = \sum_{k=0}^{\infty} \gamma^k p(s_0 \to s, k, \mu_\theta)$
- $\rho^{\mu_\theta}(s) = \int_{\mathcal{S}} p_0(s_0) \rho^{\mu_\theta}(s_0 \to s) ds_0$: discounted visitation frequency
- $d^{\mu_\theta}(s) = \frac{\rho^{\pi_\theta}(s)}{\sum_{x \in \mathcal{S}} \rho^{\pi_\theta}(x)}$: discounted stationary distribution

- some regularity conditions assumed for interchanging gradient and expectation
  - continuity w.r.t. a, θ : $p(s'|s,a)$, $r(s,a)$, $\mu_\theta(s)$
- cf. Stochastic PG

$$\nabla_\theta J(\pi_\theta) = \int_{\mathcal{S}} \rho^{\pi_\theta}(s) \int_{\mathcal{A}} \nabla_\theta \pi_\theta(a|s) Q^{\pi_\theta}(s, a)\, da\, ds \propto \mathbb{E}_{s \sim d^{\pi_\theta}, a \sim \pi_\theta}[\nabla_\theta \log \pi_\theta(a|s)\, Q^{\pi_\theta}(s, a)]$$

Implementing DPG (On-policy)

- $\nabla_\theta J(\mu_\theta) \propto \mathbb{E}_{s\sim d^\mu}\left[\nabla_\theta \mu_\theta(s) \nabla_{a=\mu_\theta(s)} Q^\mu(s,a)\right]$
  - Instead of the unknown true action value $Q^\mu(s,a)$
  - Use $Q_w(s,a)$: function approximation (such as NN)

- **Algorithm Deterministic Actor-Critic (on-policy, SARSA)**

1. initialize policy $\mu_\theta(s)$ and action value function $Q_w(s,a)$
2. for e = 1…N (# of episodes)
   - $t = 0$, sample initial state s from $p_0(s)$; $a = \mu_\theta(s)$
   - while s is not terminal or $t \le t_{\max}$
     - perform a; observe r, s'; $a' = \mu_\theta(s')$
     - update actor
       - $\theta \leftarrow \theta + \alpha_\theta \nabla_\theta \mu_\theta(s) \nabla_{b=a} Q_w(s,b)$ (Note $a = \mu_\theta(s)$)
     - update critic
       - $\delta = r + \gamma Q_w(s', a') - Q_w(s,a)$
       - $w \leftarrow w + \alpha_w \delta \nabla_w Q_w(s,a)$
     - $s \leftarrow s',\ a \leftarrow a',\ t \leftarrow t+1$

- action is selected by deterministic $\mu(s)$ →

Implementing DPG (Off-policy)

- Use (stochastic) behavioral policy $\beta(a|s)$ to ensure enough exploration
  - no importance sampling factor is required for both actor and critic
    - actor : DPG, no expectation w.r.t. action is required in DPG
      - SPG $\nabla_\theta J(\pi_\theta) \propto \mathbb{E}_{s\sim d^{\pi_\theta}, a\sim\pi_\theta}[\nabla_\theta \log \pi_\theta(a|s)\, Q^{\pi_\theta}(s,a)]$
      - DPG $\nabla_\theta J(\mu_\theta) \propto \mathbb{E}_{s\sim d^{\mu_\theta}}\left[\nabla_\theta \mu_\theta(s) \nabla_{a=\mu_\theta(s)} Q^{\mu_\theta}(s,a)\right]$
    - critic : Q-learning, no expectation w.r.t. action is required in Q-learning
      - TD target : $r + \gamma \max_{a\in\mathcal{A}} Q^{\mu_\theta}(s',a) \approx$ [ ]
        - » we are updating $\mu_\theta(s')$ s.t. $\max_{a\in\mathcal{A}} Q^{\mu_\theta}(s',a) \approx Q^{\mu_\theta}\left(s', \mu_\theta(s')\right)$

- **Algorithm Deterministic Actor-Critic (off-policy)**

1. initialize policy $\mu_\theta(s)$ and action value function $Q_w(s,a)$
2. for e = 1…N (# of episodes)
   - $t = 0$, sample initial state s from $p_0(s)$
   - while s is not terminal or $t < t_{\max}$
     - select action $a\sim\beta(a|s)$ (eg. [ ]
     - perform a; observe r, s'
     - update actor (θ update is indep. of $a\sim\beta(a|s)$ and r)
       - $\theta \leftarrow \theta + \alpha_\theta \nabla_\theta \mu_\theta(s) \nabla_{b=\mu_\theta(s)} Q_w(s,b)$
     - update critic
       - $\delta =$ [ ]
       - $w \leftarrow w + \alpha_w \delta \nabla_w Q_w(s,a)$
     - $s \leftarrow s'$, $t \leftarrow t+1$

TAQ. Can we use any β ?

[ ]

DDPG (Deep DPG, ICLR 2016 by Deepmind)

- DDPG = DQN + DPG
  - use [ ] for actor and critic
  - use [ ] for sample efficiency
  - fixed target → exponential smoothing target (soft update)

- **Algorithm DDPG**

1. initialize critic $Q_w(s,a)$ and actor $\mu_\theta(s)$
2. initialize critic target param $w' \leftarrow w$ and actor target param $\theta' \leftarrow \theta$
3. initialize replay buffer RB
4. for e = 1…M (# of episodes)
   - $t = 0$, sample initial state s from $p_0(s)$
   - while s is not terminal or $t < t_{\max}$
     - select action $a = \mu_\theta(s) +$ exploration noise (eg. $a \sim N(\mu_\theta(s), \sigma^2)$)
     - perform a and observe r, s' → store transition seg $= (s,a,r,s')$ in RB
     - sample a minibatch $MB = \{(s_i, a_i, r_i, s_i'); i = 1..N\}$ from RB
       - set target $y_i = r_i + \gamma Q_{w'}(s_i', \mu_{\theta'}(s_i'))$ for $i \in MB$
       - update critic w by min $\frac{1}{N}\sum_{i \in MB}\left(y_i - Q_w(s_i, a_i)\right)^2$
       - update actor θ by max $\frac{1}{N}\sum_{i \in MB} Q_w\left(s_i, \mu_\theta(s_i)\right)$
     - update target parameters (exponential smoothing, soft update)
       - $w' \leftarrow \tau w + (1-\tau)w'$ and $\theta' \leftarrow \tau\theta + (1-\tau)\theta'$
     - $s \leftarrow s'$, $t \leftarrow t+1$

TAQ. Why do we need θ'?

$\theta \leftarrow \theta + \alpha_\theta \frac{1}{N}\sum_i$ [ ]

TAQ. What value of τ to use?

Policy based RL (Policy Optimization)

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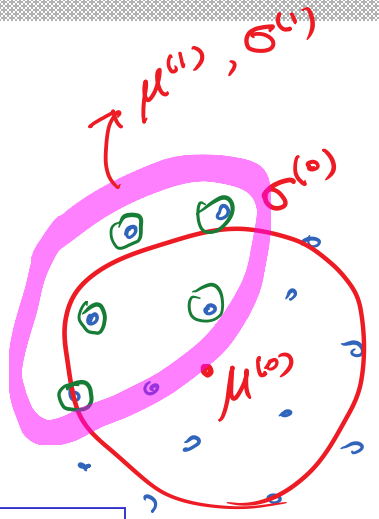
Simulation optimization (SO)

- a.k.a. "Simulation-based optimization" or "Optimization via simulation"
- SO problem

$$\min_{\theta \in \Theta} J(\theta) = \mathbb{E}_{\xi}[f(\theta, \xi)] \quad \text{(Note : } \mathbb{E}_{\xi}[f(\theta, \xi)] \neq f(\theta, \mathbb{E}[\xi])\text{)}$$

  - Characteristics of SO problem
    - $f(\theta, \xi)$: evaluated by simulation (performance measure of a system)
    - with noise ξ, whose distribution is usually unknown or uncontrollable
    - $\mathbb{E}_{\xi}[f(\theta, \xi)] \approx \frac{1}{n}\sum_{i=1}^{n} f(\theta, \xi_i)$
      - » $f(\theta, \xi_i)$: simulation output, which may take long time
    - $f(\theta, \xi)$: non-convex → many local optimal
    - $\nabla_{\theta} f(\theta, \xi)$: gradient may or may not available
- Classical approaches (derivative free methods)
  - Meta-heuristic w/ random search : GA, SA, Tabu search, …
  - Surrogate model : response surface method, GPR(or Krigging)→BO
  - Discrete Opt. via Sim. : ranking & selection
  - CEM(Cross entropy method)
- RL can be viewed as a method for SO problem
  - some RL methods exploit sequential nature of $f(\theta, \xi)$
  - Value based, Policy based, Actor-Critic

Cross Entropy Method



- Agent
  - Policy function π_θ (deterministic or stochastic)
    - θ : parameter for policy function
  - $\theta \sim P_{\mu,\Sigma}(.)$: distribution for policy param.

- **Algorithm CEM**

1. initialize distribution param. μ and Σ
2. for g = 1…N (# of generations) (or until convergence of population)
   - Sample $\theta_i \sim P_{\mu,\Sigma}(\theta)$
   - Roll-out (Perform an episode) under policy π_{θ_i} → Return R_i
   - Select elite set $E = \{e \mid R_e \in top\ p\%\}$
   - Update μ, Σ using MLE on E

$$\mu_{new}, \Sigma_{new} = \underset{\mu,\Sigma}{\operatorname{argmax}} \prod_{e \in E} P_{\mu,\Sigma}(\theta_e) = \underset{\mu,\Sigma}{\operatorname{argmax}} \sum_{e \in E} \log P_{\mu,\Sigma}(\theta_e)$$

when $P_{\mu,\Sigma}(.)$ is normal :

$$\mu_{new} = \frac{1}{|E|} \sum_{e \in E} \theta_e\,, \qquad \Sigma_{new} = \frac{1}{|E|} \sum_{e \in E} (\theta_e - \mu)(\theta_e - \mu)^T$$

3. return μ (or θ_{best} if deterministic environment)

TAQ. Why is this called 'CEM'?

Update population parameters – basic ideas

- How to update μ, Σ : basic ideas
  - Agent w/ policy π_{θ_i}, where $\theta_i \sim N(\mu, \Sigma)$ → return R_i
  - Sort w.r.t. R_i
  - Elite set $\mathrm{E} = \{\theta_i; i = 1..K_e\}$, i : rank of π_{θ_i}
  - Compute weight w_i
    - $w_i = \frac{1}{K_e}$: equal weight
    - $w_i = \frac{R_i}{\sum_{j \in E} R_j}$: return proportional weight
    - $w_i = \frac{l_i}{\sum_{j \in E} l_j}, l_i = log\frac{1+K_e}{i}$: rank based weight
  - Update mean μ
    - $\mu_{new} = \sum_{i \in E} w_i \theta_i$
  - Update sigma Σ
    - Full cov. matrix : $\Sigma_{new} = \sum_{i \in E} w_i (\theta_i - \mu_{old})(\theta_i - \mu_{old})^T + \epsilon I$
    - Diagonal matrix : $\Sigma_{new} = diag(\sigma_1^2, \ldots, \sigma_n^2) + \epsilon I$, $\sigma_c^2 = \sum_{i \in E} w_i \left(\theta_{i,c} - \mu_{old,c}\right)^2$

      $+ \epsilon I$: noise to prevent premature convergence

$\mu \xleftarrow{\alpha} \mu_{new}$

$\Sigma \xleftarrow{\alpha} \Sigma_{new}$

$x \xleftarrow{\alpha} x_{new}$

$\equiv x \leftarrow x + \alpha(x_{new} - x)$

$= (1-\alpha)x + \alpha x_{new}$

Update population parameters - using gradient

- How to update μ, σ using gradient?
  - $J(\mu,\sigma) = E_{\theta\sim f_{\mu,\sigma}(.)}[R(\theta)] = \int_\theta f_{\mu,\sigma}(\theta)R(\theta)d\theta$, $R(\theta)$: return from the env.
  - $\nabla_\mu J = \nabla_\mu \int_\theta f_{\mu,\sigma}(\theta)R(\theta)d\theta = \int_\theta \nabla_\mu f_{\mu,\sigma}(\theta)R(\theta)d\theta$

    $= \int_\theta \nabla_\mu \log f_{\mu,\sigma}(\theta)\, f_{\mu,\sigma}(\theta)R(\theta)d\theta = E_{\theta\sim f_{\mu,\sigma}(.)}\left[\nabla_\mu \log f_{\mu,\sigma}(\theta)\, R(\theta)\right]$
  - $\nabla_\sigma J = E_{\theta\sim f_{\mu,\sigma}(.)}\left[\nabla_\sigma \log f_{\mu,\sigma}(\theta)\, R(\theta)\right]$
  - $\mu_{new} = \mu_{old} + \alpha_\mu \nabla_\mu J, \quad \sigma_{new} = \sigma_{old} + \alpha_\sigma \nabla_\sigma J$
- Gaussian dist. case $f_{\mu,\sigma}(\theta) = N(\theta;\mu,\sigma)$
  - $\log f_{\mu,\sigma}(\theta) = -\log\sigma - \frac{1}{2}\log 2\pi - \frac{(\theta-\mu)^2}{2\sigma^2}$
  - $\nabla_\mu \log f_{\mu,\sigma}(\theta) = \frac{(\theta-\mu)}{\sigma^2}, \quad \nabla_\sigma \log f_{\mu,\sigma}(\theta) = -\frac{1}{\sigma} + \frac{(\theta-\mu)^2}{\sigma^3} = \frac{1}{\sigma}\left[\frac{(\theta-\mu)^2-\sigma^2}{\sigma^2}\right]$

Update population parameters – advanced ideas

- reward weighted regression
  - Dayan & Hinton, NC1997; Peters & Schaal, ICML 2007

$$\mu_{new}, \sigma_{new} = \underset{\mu,\sigma}{\operatorname{argmax}} \sum_{e \in E} R_e \log P_{\mu,\sigma}(\theta_e)$$

- policy improvement with path integral (PI^2)
  - Theodorou, Burchli, Schaal, AISTATS 2010

$$\mu_{new}, \sigma_{new} = \underset{\mu,\sigma}{\operatorname{argmax}} \sum_{e \in E} \exp(\lambda R_e) \log P_{\mu,\sigma}(\theta_e)$$

- covariance matrix adaptation
  - CMA, Hansen et al 1996; CMA-ES, Hansen et al 2003

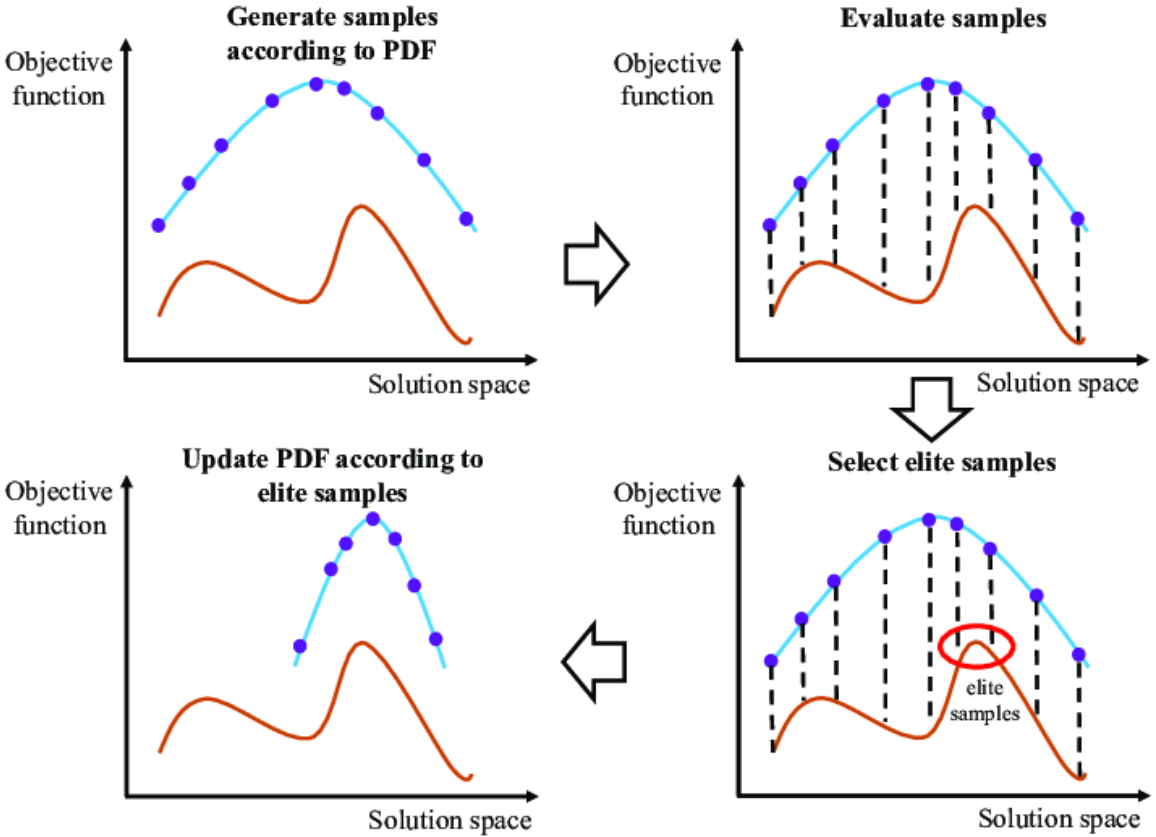
$$\mu_{new}, \sigma_{new} = \underset{\mu,\sigma}{\operatorname{argmax}} \sum_{e \in E} w_e \log P_{\mu,\sigma}(\theta_e)$$

- PoWER (Kober & Peters, NIPS 2007)

$$\mu_{new} = \mu + \alpha \sum_{e \in E} (\theta_e - \mu) R_e \Big/ \sum_{e \in E} R_e$$

References for CEM

- de Boor, PT., Kroese, D.P., Mannor, S. *et al.* A Tutorial on the Cross-Entropy Method. *Annals of OR* **134**, 19–67 (2005).
- Margolin, A. (2005). On the convergence of the cross-entropy method. *Mathematics of OR*, 30(4), 889–90
- Costa, A., Owen, J., & Kroese, D.P. (2005). Convergence properties of the cross-entropy method for discrete optimization. *Annals of Operations Research*, 137(1), 111–127



CEM for ContCartpole

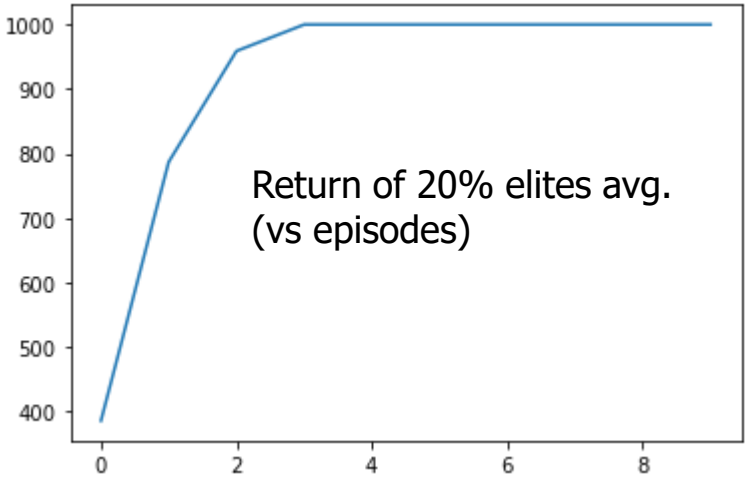
- state $s \in R^4$, $\gamma = 1$
- reward : +1 for each step (original cartpole reward)
- policy $\mu_\theta(s) = \tanh(\theta^T s)$, $\theta \in R^4$ (linear deterministic policy)
- agents : $\{\theta_i \sim N(\mu, \sigma^2); i = 1..100\}$
- 10 iterations with elite set = 20%
  - update : simple mean and var. of elite set

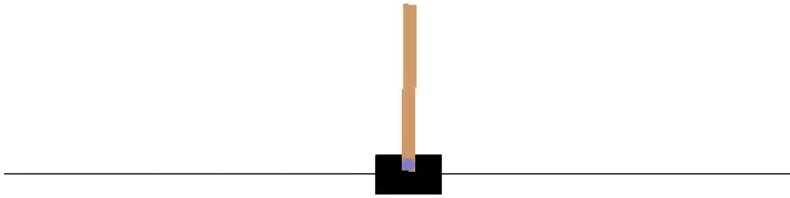
μ_0 = (0, 0, 0, 0), σ_0 = (1, 1, 1, 1)
iteration 1,
20% max, mean, min = (1000.00, 385.45, 122.40)
μ = [[0.50383628 -0.30105619 1.33247495 1.03958699]]
σ = [[0.81319874 0.64033492 0.49762544 0.44820708]]

iteration 2,
20% max, mean, min = (854.60, 786.90, 606.00)
μ = [[0.43116921 -0.54730563 1.62261229 1.12824928]]
σ = [[0.41772073 0.49221033 0.35794755 0.11910823]]

iteration 3,
20% max, mean, min = (1000.00, 958.90, 830.80)
μ = [[0.30924589 -0.13225763 1.72004417 1.05535124]]
σ = [[0.14099952 0.37946304 0.34014202 0.12380391]]

iteration 4,
20% max, mean, min = (1000.00, 1000.00, 1000.00)
μ = [[0.30976172 0.09474708 1.80581635 1.1156926]]
σ = [[0.20654872 0.29298425 0.4786936 0.15083248]]



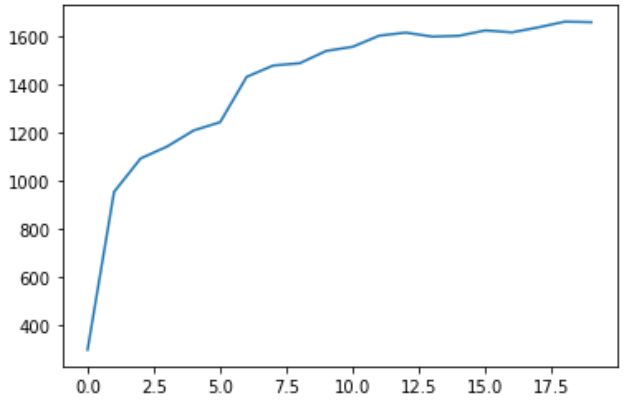


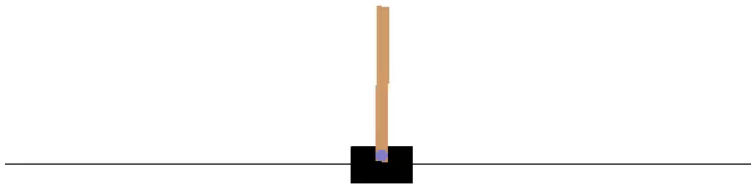
Agent trained by CEM # of steps = 1000

θ = [0.281 0.513 1.648 0.929]

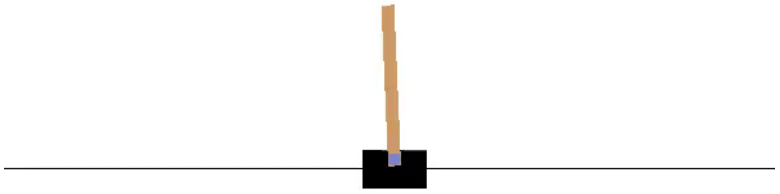
CEM for Cartpole

- Changing reward
  - default reward : +1 for each step
  - angle reward : 1+10*|pole angle| at each step
    - More slanted → more reward
    - How to maintain pole slanted without falling





Default reward (+1 for each steps)
θ = [0.281 0.513 1.648 0.929]



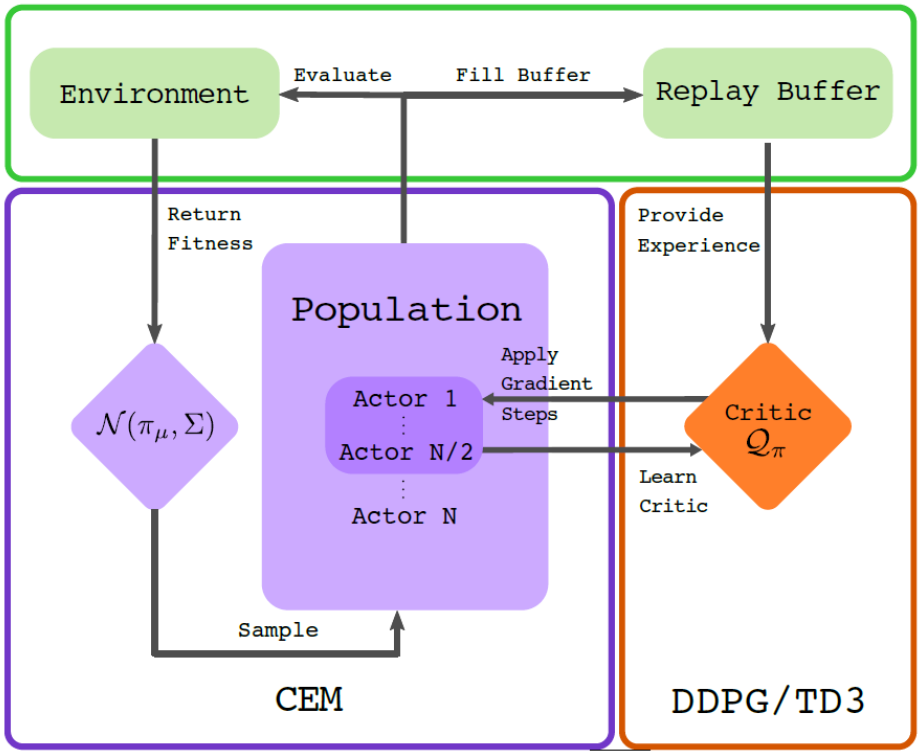
Angle reward (pole angle)
θ =[0.198 0.965 1.067 2.114]

CEM : Pros, Cons

- Pros
  - easy to implement
  - easily parallelizable
  - works quite well in many cases
  - wide applicability
    - stochastic or deterministic policy
    - continuous or discrete state/action space
    - episodic or continuing
    - …
  - no value function needed
  - no knowledge about gradient
- Cons
  - slow speed if an episode takes long time
    - population of agents
  - limited space explored if policy parameter dim is high

CEM extension ideas

- Return weighted average
- Covariance Matrix Adaptation
  - correlated parameters
- Update population dist. using gradient
- Combining CEM with PG
  - CEM-RL : CEM + DDPG or TD3 (ICLR 2019)



A1. Equivalence of policy performance objectives

$$V^{\mu}(s) = r\big(s,\mu(s)\big) + \gamma \sum_{s'} p(s'|s,\mu(s)) V^{\mu}(s') = r\big(s,\mu(s)\big) + \gamma \sum_{s'} p(s \to s', 1, \mu) V^{\mu}(s') \quad \because p(s \to s', 1, \mu) = p(s'|s,\mu(s))$$

$$= r\big(s,\mu(s)\big) + \gamma \sum_{s'} p(s \to s', 1, \mu) \left[r\big(s',\mu(s')\big) + \gamma \sum_{s''} p(s \to s'', 1, \mu) V^{\mu}(s'') \right]$$

$$= r\big(s,\mu(s)\big) + \gamma \sum_{s'} p(s \to s', 1, \mu) r\big(s',\mu(s')\big) + \gamma^2 \sum_{s''} p(s \to s'', 2, \mu) V^{\mu}(s'')$$

$$= r\big(s,\mu(s)\big) + \gamma \sum_{s'} p(s \to s', 1, \mu) r\big(s',\mu(s')\big) + \gamma^2 \sum_{s''} p(s \to s'', 2, \mu) r\big(s'',\mu(s'')\big) + \cdots$$

$$= \sum_{t=0}^{\infty} \gamma^t \sum_{x} p(s \to x, t, \mu) r\big(x,\mu(x)\big) = \sum_{x} \sum_{t=0}^{\infty} \gamma^t \, p(s \to x, t, \mu) r\big(x,\mu(x)\big)$$

$$= \sum_{x} \rho^{\mu}(s \to x) r\big(x,\mu(x)\big), \text{ where } \rho^{\mu}(s \to x) = \sum_{t=0}^{\infty} \gamma^t \, p(s \to x, t, \mu)$$

$$J(\mu) = \mathbb{E}_{s_0 \sim p_0}[V^{\mu}(s_0)] = \sum_{s_0} p_0(s_0) V^{\mu}(s_0) = \sum_{s_0} \sum_{s} p_0(s_0) \rho^{\mu}(s_0 \to s) r\big(s,\mu(s)\big)$$

$$= \sum_{x} \left[\sum_{s_0} p_0(s_0) \rho^{\mu}(s_0 \to s) \right] r\big(s,\mu(s)\big) = \sum_{s} \rho^{\mu}(s) r\big(s,\mu(s)\big) \propto \mathbb{E}_{s \sim d^{\mu}}\big[r\big(s,\mu(s)\big)\big]$$

$\rho^{\mu}(x) = \sum_{s} p_0(s_0) \rho^{\mu}(s_0 \to s) = \sum_{s} \sum_{t=0}^{\infty} \gamma^t \, p_0(s) p(s \to x, t, \mu)$: discounted visitation frequency

$d^{\mu}(s) = \dfrac{\rho^{\mu}(s)}{\sum_{x \in \mathcal{S}} \rho^{\mu}(x)}$: discounted stationary distribution

for continuing task, $\mathbb{E}_{s \sim d^{\mu}}[V^{\mu}(s)] = \frac{1}{1-\gamma} \mathbb{E}_{s \sim d^{\mu}}\big[r\big(s,\mu(s)\big)\big]$

A2. Deterministic Policy Gradient Theorem - Proof

- $\nabla_\theta V^{\mu_\theta}(s) = \nabla_\theta Q^{\mu_\theta}\big(s, \mu_\theta(s)\big)$

$$= \nabla_\theta \left(r\big(s, \mu_\theta(s)\big) + \int_{\mathcal{S}} \gamma p\big(s' | s, \mu_\theta(s)\big) V^{\mu_\theta}(s') ds' \right)$$ Bellman Equation

$$= \nabla_\theta \mu_\theta(s) \nabla_{a=\mu_\theta(s)} r(s,a) + \nabla_\theta \int_{\mathcal{S}} \gamma p\big(s' | s, \mu_\theta(s)\big) V^{\mu_\theta}(s') ds'$$

$$= \nabla_\theta \mu_\theta(s) \nabla_{a=\mu_\theta(s)} r(s,a) + \int_{\mathcal{S}} \nabla_\theta \gamma p\big(s' | s, \mu_\theta(s)\big) V^{\mu_\theta}(s') ds'$$ exchange ∇_θ and $\int_{\mathcal{S}}$

$$= \nabla_\theta \mu_\theta(s) \nabla_{a=\mu_\theta(s)} r(s,a) + \int_{\mathcal{S}} \gamma \left(p\big(s' | s, \mu_\theta(s)\big) \nabla_\theta V^{\mu_\theta}(s') + \nabla_\theta \mu_\theta(s) \nabla_{a=\mu_\theta(s)} p(s' | s, a) V^{\mu_\theta}(s') \right) ds'$$

$$= \nabla_\theta \mu_\theta(s) \nabla_{a=\mu_\theta(s)} \left(r(s,a) + \int_{\mathcal{S}} \gamma p(s' | s, a) V^{\mu_\theta}(s') ds' \right) + \int_{\mathcal{S}} \gamma p\big(s' | s, \mu_\theta(s)\big) \nabla_\theta V^{\mu_\theta}(s) ds'$$

$$= \nabla_\theta \mu_\theta(s) \nabla_{a=\mu_\theta(s)} Q^{\mu_\theta}(s,a) + \int_{\mathcal{S}} \gamma p\big(s' | s, \mu_\theta(s)\big) \nabla_\theta V^{\mu_\theta}(s) ds'$$

$$= \nabla_\theta \mu_\theta(s) \nabla_{a=\mu_\theta(s)} Q^{\mu_\theta}(s,a) + \gamma \int_{\mathcal{S}} p(s \to s', 1, \mu_\theta) \nabla_\theta V^{\mu_\theta}(s') ds'$$

$$= \nabla_\theta \mu_\theta(s) \nabla_{a=\mu_\theta(s)} Q^{\mu_\theta}(s,a)$$
$$+ \gamma \int_{\mathcal{S}} p(s \to s', 1, \mu_\theta) \left(\nabla_\theta \mu_\theta(s') \nabla_{a=\mu_\theta(s')} Q^{\mu_\theta}(s', a) + \gamma \int_{\mathcal{S}} p(s' \to s'', 1, \mu_\theta) \nabla_\theta V^{\mu_\theta}(s'') ds'' \right) ds'$$

expand using

$p(s \to s', k, \mu_\theta)$: state transition prob. from s to s' after k steps

A2. Deterministic Policy Gradient Theorem - Proof

$$\nabla_\theta V^{\mu_\theta}(s) = \nabla_\theta \mu_\theta(s) \nabla_{a=\mu_\theta(s)} Q^{\mu_\theta}(s,a) + \gamma \int_{\mathcal{S}} p(s \to s', 1, \mu_\theta) \nabla_\theta \mu_\theta(s') \nabla_{a=\mu_\theta(s')} Q^{\mu_\theta}(s',a) ds'$$

$$+\gamma^2 \int_{\mathcal{S}} \int_{\mathcal{S}} p(s \to s', 1, \mu_\theta) p(s' \to s'', 1, \mu_\theta) \nabla_\theta V^{\mu_\theta}(s'') ds'' ds'$$

change order

$$\int_{\mathcal{S}} p(s \to s', 1, \mu_\theta) p(s' \to s'', 1, \mu_\theta) ds' = p(s \to s'', 2, \mu_\theta)$$

$$= \nabla_\theta \mu_\theta(s) \nabla_{a=\mu_\theta(s)} Q^{\mu_\theta}(s,a) + \gamma \int_{\mathcal{S}} p(s \to s', 1, \mu_\theta) \nabla_\theta \mu_\theta(s') \nabla_{a=\mu_\theta(s')} Q^{\mu_\theta}(s',a) ds' + \gamma^2 \int_{\mathcal{S}} p(s \to s'', 2, \mu_\theta) \nabla_\theta V^{\mu_\theta}(s'') ds''$$

$$= \nabla_\theta \mu_\theta(s) \nabla_{a=\mu_\theta(s)} Q^{\mu_\theta}(s,a) + \gamma \int_{\mathcal{S}} p(s \to s', 1, \mu_\theta) \nabla_\theta \mu_\theta(s') \nabla_{a=\mu_\theta(s')} Q^{\mu_\theta}(s',a) ds'$$

$$+\gamma^2 \int_{\mathcal{S}} p(s \to s'', 2, \mu_\theta) \nabla_\theta \mu_\theta(s'') \nabla_{a=\mu_\theta(s'')} Q^{\mu_\theta}(s'',a) ds'' + \gamma^3 \int_{\mathcal{S}} p(s \to s''', 3, \mu_\theta) \nabla_\theta V^{\mu_\theta}(s''') ds''' = \cdots$$

$$= \int_{\mathcal{S}} \sum_{k=0}^{\infty} \gamma^k p(s \to x, k, \mu_\theta) \nabla_\theta \mu_\theta(x) \nabla_{a=\mu_\theta(x)} Q^{\mu_\theta}(x,a) \, dx = \int_{\mathcal{S}} \rho^{\mu_\theta}(s \to x) \nabla_\theta \mu_\theta(x) \nabla_{a=\mu_\theta(x)} Q^{\mu_\theta}(x,a) dx$$

$$\nabla_\theta J(\mu_\theta) = \nabla_\theta \int_{\mathcal{S}} p_0(s) V^{\mu_\theta}(s) ds = \int_{\mathcal{S}} p_0(s) \nabla_\theta V^{\mu_\theta}(s) ds$$

$$= \int_{\mathcal{S}} \left[\int_{\mathcal{S}} p_0(s) \rho^{\mu_\theta}(s \to x) \, ds \right] \nabla_\theta \mu_\theta(x) \nabla_{a=\mu_\theta(x)} Q^{\mu_\theta}(x,a) dx$$

$$= \int_{\mathcal{S}} \rho^{\mu_\theta}(x) \nabla_\theta \mu_\theta(x) \nabla_{a=\mu_\theta(x)} Q^{\mu_\theta}(x,a) \, dx$$

$$= Z \int_{\mathcal{S}} d^{\mu_\theta}(x) \nabla_\theta \mu_\theta(x) \nabla_{a=\mu_\theta(x)} Q^{\mu_\theta}(x,a) \, dx$$

$$\propto \mathbb{E}_{x \sim d^{\mu_\theta}} \left[\nabla_\theta \mu_\theta(x) \nabla_{a=\mu_\theta(x)} Q^{\mu_\theta}(x,a) \right]$$

$$\rho^{\mu_\theta}(s \to x) = \sum_{k=0}^{\infty} \gamma^k p_0(s) p(s \to x, k, \mu_\theta) dx$$

$$\rho^{\mu_\theta}(x) = \int_{\mathcal{S}} p_0(s) \rho^{\mu_\theta}(s \to x) ds$$

: discounted visitation frequency

$$d^{\mu_\theta}(x) = \frac{\rho^{\pi_\theta}(x)}{Z}, \qquad Z = \sum_{s \in \mathcal{S}} \rho^{\pi_\theta}(s)$$

: discounted stationary distribution

A3. Off-Policy Stochastic PG

- Off-Policy Actor-Critic
  - Target policy: $\pi_\theta(a|s)$
  - Behavior policy: $\beta(a|s)$
  - $J_\beta(\pi_\theta) = \mathbb{E}_{s\sim d^\beta}[V^{\pi_\theta}(s)] = \mathbb{E}_{s\sim d^\beta, a\sim\pi_\theta}[Q^{\pi_\theta}(s,a)] = \mathbb{E}_{s\sim d^\beta}\left[\int_S \pi_\theta(a|s) Q^{\pi_\theta}(s,a)\, da\right]$
  - $\nabla_\theta J_\beta(\pi_\theta) = \mathbb{E}_{s\sim d^\beta}\left[\int_S \nabla_\theta \pi_\theta(a|s) Q^{\pi_\theta}(s,a) + \pi_\theta(a|s)\nabla_\theta Q^{\pi_\theta}(s,a)\, ds\right]$

$$\approx \mathbb{E}_{s\sim d^\beta}\left[\int_S \nabla_\theta \pi_\theta(a|s) Q^{\pi_\theta}(s,a)\, ds\right] \quad \langle \text{ignore} \nabla_\theta Q^\pi(s,a), \text{Degris et al, ICML '12} \rangle$$

$$= \mathbb{E}_{s\sim d^\beta}\left[\int_S \pi_\theta(a|s) \frac{\nabla_\theta \pi_\theta(a|s)}{\pi_\theta(a|s)} Q^{\pi_\theta}(s,a)\, ds\right] \quad \langle \text{score function technique} \rangle$$

$$= \mathbb{E}_{s\sim d^\beta}\left[\int_S \pi_\theta(a|s) \nabla_\theta \log \pi_\theta(a|s)\, Q^{\pi_\theta}(s,a)\, ds\right]$$

$$= \mathbb{E}_{s\sim d^\beta,\, a\sim\pi_\theta}\left[\nabla_\theta \log \pi_\theta(a|s)\, Q^{\pi_\theta}(s,a)\right] \quad \langle \text{hard to use samples from } \beta \rangle$$

$$= \mathbb{E}_{s\sim d^\beta}\left[\int_S \beta(a|s) \frac{\pi_\theta(a|s)}{\beta(a|s)} \nabla_\theta \log \pi_\theta(a|s)\, Q^{\pi_\theta}(s,a)\, ds\right] \langle \text{importance sampling} \rangle$$

$$= \mathbb{E}_{s\sim d^\beta, a\sim\beta}\left[\frac{\pi_\theta(a|s)}{\beta(a|s)} \nabla_\theta \log \pi_\theta(a|s)\, Q^{\pi_\theta}(s,a)\right] \quad \langle \text{use samples from } \beta \rangle$$